

PHYSICS

1. What is the dimensional formula of $\frac{1}{\mu_0 \epsilon_0}$ (where μ_0 is permeability and ϵ_0 is permittivity of free space)

- (1) LT^{-1}
 (2) L^2T^{-2}
 (3) MLT^{-1}
 (4) ML^2T^{-2}

Sol. $C = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$

$$C^2 = \frac{1}{\mu_0 \epsilon_0}$$

$$[C^2] = L^2T^{-2}$$

2. An electron with initial velocity $\vec{v} = v_0 \hat{i}$ is projected in magnetic field $\vec{B} = v_0 \hat{j}$. If the de-broglie wavelength of e^- is λ_0 after time 't'. Find the initial de-broglie wavelength.

Sol. $\lambda_0 = \frac{h}{p}$

3. An equilateral prism is made of a material of refractive index $\sqrt{2}$ find angle of incidence for minimum deviation of the light ray.

- (1) 60°
 (2) 30°
 (3) 37°
 (4) 45°

Sol. $1 \sin i = \sqrt{2} \sin 30^\circ$

$$\sin i = \sqrt{2} \times \frac{1}{2}$$

$$= \frac{1}{\sqrt{2}}$$

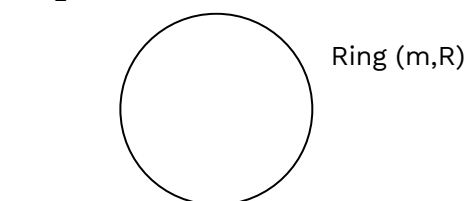
$$i = 45^\circ$$

4. Calculate the MOI of a ring of mass m and radius R about an axis tangent to the ring and in the plane of ring

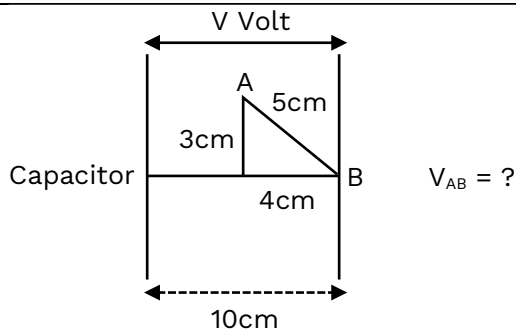
Sol. BY using parallel axis theorem

$$I = \frac{mR^2}{2} + mR^2$$

$$I = \frac{3}{2}mR^2$$

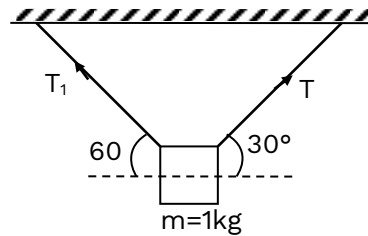
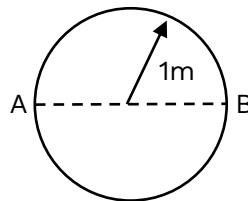


5.

Sol. $10 \rightarrow V_0$

$$1 \rightarrow \frac{V_0}{10}$$

$$4 \text{ cm} \rightarrow \frac{V_0}{10} \times 4 = \frac{2V_0}{5}$$

6. Find the tensions T_1 and T_2 Sol. $T_1 = 5\sqrt{3}$
 $T_2 = 5$ 7. A particle moves on a circular path of radius 1 m. Find its displacement when it moves from $A \rightarrow B \rightarrow A \rightarrow B$. Also its distance are it moves from $A \rightarrow B \rightarrow A \rightarrow B \rightarrow A$.Sol. $A \rightarrow B \rightarrow A \rightarrow B$
 $S = 2\pi R + \pi R = 3\pi R$
 $\vec{S} = 2R$
 $\frac{S}{S} = \frac{3\pi R}{2R} = \frac{3\pi}{2}$ 8. A particle of mass m is projected at angle 60° with horizontal. If initial kinetic energy is KE_0 and kinetic energy at maximum height is $\frac{KE_0}{x}$. Find the value of x .Sol. $\frac{1}{2}mv^2 = KE_0$
 $\frac{1}{2}m\left(\frac{u}{2}\right)^2 = KE = \frac{1}{4}\left(\frac{1}{2}mv^2\right) = \frac{KE}{4}$

9. $\left. \begin{array}{l} u = 18 \text{ cm} \\ R = 12 \text{ cm} \end{array} \right\} \Rightarrow \text{Convex \& concave mirror}$

$$\frac{h_{i(\text{concave})}}{h_{i(\text{convex})}} = ?$$

10. A satellite is nine times closer to earth compared to moon. Time period of moon is 27 days then period of satellite is
 (A) 1 days
 (B) 3 days
 (C) 9 day
 (D) $3\sqrt{3}$ days

Sol. $T^2 \propto r^3$

$$\frac{(27)^2}{T^2} = \frac{(r)^3}{(r/9)^3} \Rightarrow \frac{(3^6)}{T^2} = 9^3$$

$$3^6 = 3^6 T^2$$

$$T^2 = 1 \Rightarrow T = 1$$

11. Two identical drops of radius R and surface tension 'T' coalesce to form a bigger drop. The change in surface energy in this process is

(1) $4\pi R^2 T \left[1 - 2^{-\frac{1}{3}} \right]$

(2) $8\pi R^2 T \left[1 + 2^{\frac{1}{3}} \right]$

(3) $4\pi R^2 T \left[1 + 2^{\frac{1}{3}} \right]$

(4) $8\pi R^2 T \left[2^{-\frac{1}{3}} - 1 \right]$

Sol. $U_i = 4\pi R^2 S \times 2$
 $U_f = 4\pi 2^{2/3} R^2 S$
 $U_f - U_i = 4\pi R^2 S \{2^{2/3} - 2\}$
 $= 8\pi R^2 S \{2^{-1/3} - 1\}$

12. If force (F) = 5N then length is 1.4m and if force (F) = 7N, then length is 1.56m. Find the original length.

Sol. $5 = K(1.4 - \ell_0) \quad \dots(1)$

$7 = K(1.56 - \ell_0) \quad \dots(2)$

$$\frac{5}{7} = \frac{1.4 - \ell_0}{1.56 - \ell_0}$$

$$5[1.56 - \ell_0] = 7[1.4 - \ell_0]$$

$$5 \times 1.56 - 7 \times 1.4 = 5\ell_0 - 7\ell_0$$

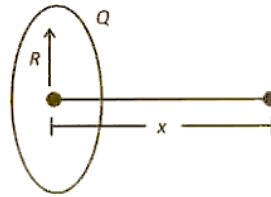
13. Figure shows a uniformly charged ring having charge Q and radius R . Find the distance from the centre on the axis of the ring where electric field is maximum

(1) $R\sqrt{2}$

(2) $\frac{R}{R^2}$

(3) $2R$

(4) R

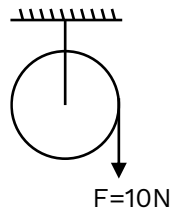


Sol. $E = \frac{KQx}{(R^2 + x^2)^{3/2}}$

$$\frac{dE}{dx} = 0$$

$$X = \frac{R}{\sqrt{2}}$$

14. A disc of mass M and radius $2m$ is hinged keeping axis horizontal. If the angular acceleration of the disc is 2 rad/sec^2 . Find moment of inertia.



Sol. $10 \times 2 = I \cdot 2$
 $I = 10$

15. A solenoid having area A and length ' l ' is filled with a material having relative permeability 2. The magnetic energy stored in the solenoid is

(1) $\frac{B^2 Al}{\mu_0}$

(2) $\frac{B^2 Al}{4\mu_0}$

(3) $B^2 Al$

(4) $\frac{B^2 Al}{2\mu_0}$

Sol. $u = \frac{1}{2} LI^2$

$$L = \mu_0 n^2 (A \ell)$$

$$u = \frac{2}{2} \mu_0 n^2 A \ell \cdot I^2$$

16. $R_1 = 5\Omega$, $A_1 = 3.6 \times 10^{-3}$, $B_1 = 0.25 \text{ T}$
 $R_2 = 7\Omega$, $A_2 = 1.8 \times 10^{-3}$, $B_2 = 0.5 \text{ T}$

Torsional constant are equal then find the ratio of voltage sensitivity.

Sol. $C\theta = NIAB$

$$= \frac{Q}{V} = \frac{Q}{R \cdot I} = \frac{NAB}{CR}$$

$$\frac{Q_1}{V_1} = \frac{NA_1B_1}{CR_1}, \frac{Q_2}{V_2} = \frac{NA_2B_2}{CR_2}$$

$$\frac{A_1B_1}{A_2B_2} \times \frac{R_2}{R_1}$$

$$2 \times \frac{1}{2} \times \frac{7}{5} = 1.4$$

17. Match the List-I with the List-II:

- | | |
|-----------------------------|--------------------------------------|
| (i) Heat capacity | (a) $\text{J kg}^{-1} \text{K}^{-1}$ |
| (ii) Specific heat capacity | (b) J K^{-1} |
| (iii) Latent heat | (c) $\text{W m}^{-1} \text{K}^{-1}$ |
| (iv) Thermal conductivity | (d) J kg^{-1} |

- (1) (i)-(b), (ii)-(d), (iii)-(c), (iv)-(a)
 (2) (i)-(b), (ii)-(a), (iii)-(c), (iv)-(a)
 (3) (i)-(b), (ii)-(c), (iii)-(d), (iv)-(a)
 (4) (i)-(b), (ii)-(a), (iii)-(d), (iv)-(c)

18. Biconvex lens $R = \frac{1}{6} \text{cm}$, which of the following combination of radii can be possible to yield same power.

(1) $R_1 = \frac{1}{7}, R_2 = \frac{1}{5}$

(2) $R_1 = \frac{1}{4}, R_2 = \frac{1}{3}$

(3) $R_1 = \frac{1}{5}, R_2 = \frac{1}{3}$

(4) $R_1 = \frac{1}{6}, R_2 = \frac{1}{4}$

Sol. $\Rightarrow \frac{1}{F} = \left(\frac{3}{2} - 1\right) \left(\frac{2}{R}\right)$

$$\frac{1}{F} = \frac{1}{2} \times \frac{2}{6} = \frac{1}{6}$$

$$\frac{1}{F} = \left(\frac{3}{2} - 1\right) \left(\frac{1}{4} + \frac{1}{3}\right)$$

$$= \frac{1}{2} \left[\frac{3+4}{12} \right]$$

$$\frac{1}{F} = \frac{7}{24}$$

$$\frac{1}{F} = \left(\frac{3}{2} - 1\right) \left(\frac{1}{6} + \frac{1}{4}\right) = \frac{1}{2} \left[\frac{3+2}{12} \right] = \frac{5}{24}$$

$$\frac{1}{F} = \left(\frac{3}{2} - 1\right) \left(\frac{1}{5} + \frac{1}{3}\right)$$

$$= \frac{1}{2} \left[\frac{3+5}{15} \right]$$

$$= \frac{1}{F} = \frac{4}{15}$$

- 19.** A wave travels 1.2 cm in 0.3 sec. if the wavelength of wave is 7.5 cm and amplitude is 2 cm. Find the equation of wave assuming crest $x = 0$ at $t = 0$.

Sol. $v_{\omega} = \frac{1.2}{0.3} = 4 \text{ cm}$

$$\lambda = 7.5 \text{ cm}, A = 2 \text{ cm}$$

$$y = A \sin\left(\omega t + \frac{\pi}{2}\right)$$

$$A \sin \left[\omega \left(t - \frac{x}{v} \right) + \frac{\pi}{2} \right]$$

$$A \sin \left[\omega t - kx + \frac{\pi}{2} \right]$$

$$\omega = 2\pi F, v_{\omega} = \lambda F$$

$$4 = 7.5F$$

- 20.** If the Bohr atomic radius is A_0 . Then find the radius in ground state of Li^{2+} ion.

Sol. $R \propto \frac{n^2}{Z}$

$$Z = 3$$

$$R = \frac{A_0}{3}$$

- 21. Assertion:** In absence of external electric field. The dipole moment of an isotropic dielectric material is non-zero.

Reason: In absence of external electric field the orientation of dipole is not fixed.

- 22.** Find the kinetic energy of a satellite of mass 1000 kg at a height of 270 km from the surface of earth.

Sol. $K.E = \frac{1}{2}mv^2 = \frac{1}{2} \times 1000 \times \frac{G.M}{6670 \times 10^3}$

$$v_0 = \sqrt{gr} = \sqrt{\frac{GM}{r}}$$